will increase. These shares should be transacted as a pair to maintain risk and to minimize market impact cost. In the latter scenario it would not be advisable to submit these orders into a dark pool for execution.

The general technique to determine our program-block decomposition is through min-max optimization. That is, we seek to minimize the maximum residual risk position. This is determined as follows:

Let,

$R = (r_1, \cdots, r_k, \cdots, r_m)'$ represent the current trade portfolio

$M = \sum_{i=1}^{m} r_i$ represents the total number of shares in the portfolio

$Y = (y_1, \cdots, y_k, \cdots, y_m)'$ represents the block subset

$R - Y = (r_1 - y_1, \cdots, r_k - y_k, \cdots, r_m - y_m)'$ represents the program subset

C = one-sided covariance matrix scaled for a trading period and expressed in $(\$/\text{share})^2$

Next, let,

$Z = (z_1, \cdots, z_k, \cdots, z_m)'$ represent the untraded shares from the block subset after submission to the dark pool. That is, Y is entered into a dark pool where some trades occur. Z represents those shares that did not transact in the dark pool. Then,

$R - Y + Z = (r_1 - y_1 + z_1, \cdots, r_k - y_k + z_k, \cdots, r_m - y_m + z_m)'$ represents the residual portfolio after submission to the dark pool.

Then, the resulting total residual risk is:

$$Risk = \sqrt{(R - Y + Z)'C(R - Y + Z)} \tag{9.30}$$

with $0 \le z_k \le y_k \le r_k$ to ensure that there is no overtrading.

Mathematically, program-block decomposition can be formulated as a min-max optimization problem where we minimize the worse-case scenario. For a given number of shares S, where $S = y_1 + \cdots + y_m$, the block subset is determined as follows:

$$\underset{x}{Min}\ \underset{y}{Max} \quad (R - Y + Z)'C(R - Y + Z) \tag{9.31}$$

Subject to:

$0 \le z_k \le y_k \le r_k$	*for all k*
$S = y_1 + y_2 + \cdots + y_m$	*for all k*